Chapter 10

Differentiation of functions

10.1 Basic definitions

We can now begin the rigorous treatment of calculus in earnest, starting with the notion of a derivative. We can now define derivatives analytically, using limits, in contrast to the geometric definition of derivatives, which uses tangents. The advantage of working analytically is that (a) we do not need to know the axioms of geometry, and (b) these definitions can be modified to handle functions of several variables, or functions whose values are vectors instead of scalar. Furthermore, one's geometric intuition becomes difficult to rely on once one has more than three dimensions in play. (Conversely, one can use one's experience in analytic rigour to extend one's geometric intuition to such abstract settings; as mentioned earlier, the two viewpoints complement rather than oppose each other.)

Definition 10.1.1 (Differentiability at a point)**.** Let X be a subset of $\mathbf{R}$, and let $x_0 \in X$ be an element of X which is also a limit point of X. Let $f : X \to \mathbf{R}$ be a function. If the limit

$$\lim_{x \to x_0; x \in X - \{x_0\}} \frac{f(x) - f(x_0)}{x - x_0}$$

converges to some real number L, then we say that f is *differentiable at* x_0 *on* X *with derivative* L, and write $f'(x_0) := L$. If the limit does not exist, or if x_0 is not an element of X or not a limit point of X, we leave $f'(x_0)$ undefined, and say that f is *not differentiable at* x_0 *on* X.

Remark 10.1.2. Note that we need x_0 to be a limit point in order for x_0 to be adherent to $X - \{x_0\}$, otherwise the limit

$$\lim_{x \to x_0; x \in X - \{x_0\}} \frac{f(x) - f(x_0)}{x - x_0}$$

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would automatically be undefined. In particular, we do not define the derivative of a function at an isolated point; for instance, if one restricts the function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) := x^2$ to the domain $X := [1, 2] \cup \{3\}$, then the restriction of the function ceases to be differentiable at 3. (See however Exercise 10.1.1 below.) In practice, the domain X will almost always be an interval, and so by Lemma 9.1.21 all elements x_0 of X will automatically be limit points and we will not have to care much about these issues.

Example 10.1.3. Let $f : \mathbf{R} \to \mathbf{R}$ be the function $f(x) := x^2$, and let x_0 be any real number. To see whether f is differentiable at x_0 on $\mathbf{R}$, we compute the limit

$$\lim_{x \to x_0; x \in \mathbf{R} - \{x_0\}} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \to x_0; x \in \mathbf{R} - \{x_0\}} \frac{x^2 - x_0^2}{x - x_0}.$$

We can factor the numerator as $(x^2 - x_0^2) = (x - x_0)(x + x_0)$. Since $x \in \mathbf{R} - \{x_0\}$, we may legitimately cancel the factors of $x - x_0$ and write the above limit as

$$\lim_{x \to x_0; x \in \mathbf{R} - \{x_0\}} x + x_0$$

which by limit laws is equal to $2x_0$. Thus the function $f(x)$ is differentiable at x_0 and its derivative there is $2x_0$.

Remark 10.1.4. This point is trivial, but it is worth mentioning: if $f : X \to \mathbf{R}$ is differentiable at x_0, and $g : X \to \mathbf{R}$ is equal to f (i.e., $g(x) = f(x)$ for all $x \in X$), then g is also differentiable at x_0 and $g'(x_0) = f'(x_0)$ (why?). However, if two functions f and g merely have the same *value* at x_0, i.e., $g(x_0) = f(x_0)$, this does not imply that $g'(x_0) = f'(x_0)$. (Can you see a counterexample?) Thus there is a big difference between two functions being equal on their whole domain, and merely being equal at one point.

Remark 10.1.5. One sometimes writes $\frac{df}{dx}$ instead of f'. This notation is of course very familiar and convenient, but one has to be a little careful, because it is only safe to use as long as x is the only variable used to represent the input for f; otherwise one can get into all sorts of trouble. For instance, the function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) := x^2$ has derivative $\frac{df}{dx} = 2x$, but the function $g : \mathbf{R} \to \mathbf{R}$ defined by $g(y) := y^2$ would seem to have derivative $\frac{dg}{dx} = 0$ if y and x are independent

variables, despite the fact that g and f are exactly the same function. Because of this possible source of confusion, we will refrain from using the notation $\frac{df}{dx}$ whenever it could possibly lead to confusion. (This confusion becomes even worse in the calculus of several variables, and the standard notation of $\frac{\partial f}{\partial x}$ can lead to some serious ambiguities. There are ways to resolve these ambiguities, most notably by introducing the notion of differentiation along vector fields, but this is beyond the scope of this text.)

Example 10.1.6. Let $f : \mathbf{R} \to \mathbf{R}$ be the function $f(x) := |x|$, and let $x_0 = 0$. To see whether f is differentiable at 0 on $\mathbf{R}$, we compute the limit

$$\lim_{x \to 0; x \in \mathbf{R} - \{0\}} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0; x \in \mathbf{R} - \{0\}} \frac{|x|}{x}.$$

Now we take left limits and right limits. The right limit is

$$\lim_{x \to 0; x \in (0,\infty)} \frac{|x|}{x} = \lim_{x \to 0; x \in (0,\infty)} \frac{x}{x} = \lim_{x \to 0; x \in (0,\infty)} 1 = 1,$$

while the left limit is

$$\lim_{x \to 0; x \in (-\infty,0)} \frac{|x|}{x} = \lim_{x \to 0; x \in (0,\infty)} \frac{-x}{x} = \lim_{x \to 0; x \in (0,\infty)} -1 = -1,$$

and these limits do not match. Thus $\lim_{x \to 0; x \in \mathbf{R} - \{0\}} \frac{|x|}{x}$ does not exist, and f is not differentiable at 0 on $\mathbf{R}$. However, if one restricts f to $[0, \infty)$, then the restricted function $f|_{[0,\infty)}$ *is* differentiable at 0 on $[0, \infty)$, with derivative 1:

$$\lim_{x \to 0; x \in [0,\infty) - \{0\}} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0; x \in (0,\infty)} \frac{|x|}{x} = 1.$$

Similarly, when one restricts f to $(-\infty, 0]$, the restricted function $f|_{(-\infty,0]}$ is differentiable at 0 on $(-\infty, 0]$, with derivative -1. Thus even when a function is not differentiable, it is sometimes possible to restore the differentiability by restricting the domain of the function.

If a function is differentiable at x_0, then it is approximately linear near x_0:

Proposition 10.1.7 (Newton's approximation)**.** *Let X be a subset of $\mathbf{R}$, let $x_0 \in X$ be a limit point of X, let $f : X \to \mathbf{R}$ be a function,*

and let L be a real number. Then the following statements are logically equivalent:

(a) *f is differentiable at x_0 on X with derivative L.*

(b) *For every $\varepsilon > 0$, there exists a $\delta > 0$ such that $f(x)$ is $\varepsilon|x - x_0|$-close to $f(x_0) + L(x - x_0)$ whenever $x \in X$ is δ-close to x_0, i.e., we have*

$$|f(x) - (f(x_0) + L(x - x_0))| \leq \varepsilon|x - x_0|$$

whenever $x \in X$ and $|x - x_0| \leq \delta$.

Remark 10.1.8. Newton's approximation is of course named after the great scientist and mathematician Isaac Newton (1642–1727), one of the founders of differential and integral calculus.

Proof. See Exercise 10.1.2. □

Remark 10.1.9. We can phrase Proposition 10.1.7 in a more informal way: if f is differentiable at x_0, then one has the approximation $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$, and conversely.

As the example of the function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) := |x|$ shows, a function can be continuous at a point without being differentiable at that point. However, the converse is true:

Proposition 10.1.10 (Differentiability implies continuity)**.** *Let X be a subset of $\mathbf{R}$, let $x_0 \in X$ be a limit point of X, and let $f : X \to \mathbf{R}$ be a function. If f is differentiable at x_0, then f is also continuous at x_0.*

Proof. See Exercise 10.1.3. □

Definition 10.1.11 (Differentiability on a domain)**.** Let X be a subset of $\mathbf{R}$, and let $f : X \to \mathbf{R}$ be a function. We say that f is *differentiable on X* if, for every limit point $x_0 \in X$, the function f is differentiable at x_0 on X.

From Proposition 10.1.10 and the above definition we have an immediate corollary:

Corollary 10.1.12. *Let X be a subset of $\mathbf{R}$, and let $f : X \to \mathbf{R}$ be a function which is differentiable on X. Then f is also continuous on X.*

Now we state the basic properties of derivatives which you are all familiar with.

Theorem 10.1.13 (Differential calculus)**.** *Let X be a subset of $\mathbf{R}$, let $x_0 \in X$ be a limit point of X, and let $f : X \to \mathbf{R}$ and $g : X \to \mathbf{R}$ be functions.*

(*a*) *If f is a constant function, i.e., there exists a real number c such that $f(x) = c$ for all $x \in X$, then f is differentiable at x_0 and $f'(x_0) = 0$.*

(*b*) *If f is the identity function, i.e., $f(x) = x$ for all $x \in X$, then f is differentiable at x_0 and $f'(x_0) = 1$.*

(*c*) (*Sum rule*) *If f and g are differentiable at x_0, then $f + g$ is also differentiable at x_0, and $(f + g)'(x_0) = f'(x_0) + g'(x_0)$.*

(*d*) (*Product rule*) *If f and g are differentiable at x_0, then fg is also differentiable at x_0, and $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.*

(*e*) *If f is differentiable at x_0 and c is a real number, then cf is also differentiable at x_0, and $(cf)'(x_0) = cf'(x_0)$.*

(*f*) (*Difference rule*) *If f and g are differentiable at x_0, then $f - g$ is also differentiable at x_0, and $(f - g)'(x_0) = f'(x_0) - g'(x_0)$.*

(*g*) *If g is differentiable at x_0, and g is non-zero on X (i.e., $g(x) \neq 0$ for all $x \in X$), then $1/g$ is also differentiable at x_0, and $(\frac{1}{g})'(x_0) = -\frac{g'(x_0)}{g(x_0)^2}$.*

(*h*) *(Quotient rule) If f and g are differentiable at x_0, and g is non-zero on X, then f/g is also differentiable at x_0, and*

$$(\frac{f}{g})'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

Remark 10.1.14. The product rule is also known as the *Leibniz rule*, after Gottfried Leibniz (1646–1716), who was the other founder of differential and integral calculus besides Newton.

Proof. See Exercise 10.1.4. □

As you are well aware, the above rules allow one to compute many derivatives easily. For instance, if $f : \mathbf{R} - \{1\} \to \mathbf{R}$ is the function $f(x) := \frac{x-2}{x-1}$, then it is easy to use the above rules to show that $f'(x_0) = \frac{1}{(x_0-1)^2}$ for all $x_0 \in \mathbf{R} - \{1\}$. (Why? Note that every point x_0 in $\mathbf{R} - \{1\}$ is a limit point of $\mathbf{R} - \{1\}$.)

Another fundamental property of differentiable functions is the following:

Theorem 10.1.15 (Chain rule)**.** *Let X, Y be subsets of $\mathbf{R}$, let $x_0 \in X$ be a limit point of X, and let $y_0 \in Y$ be a limit point of Y. Let $f : X \to Y$ be a function such that $f(x_0) = y_0$, and such that f is differentiable at x_0. Suppose that $g : Y \to \mathbf{R}$ is a function which is differentiable at y_0. Then the function $g \circ f : X \to \mathbf{R}$ is differentiable at x_0, and*

$$(g \circ f)'(x_0) = g'(y_0) f'(x_0).$$

Proof. See Exercise 10.1.7. □

Example 10.1.16. If $f : \mathbf{R} - \{1\} \to \mathbf{R}$ is the function $f(x) := \frac{x-2}{x-1}$, and $g : \mathbf{R} \to \mathbf{R}$ is the function $g(y) := y^2$, then $g \circ f(x) = (\frac{x-2}{x-1})^2$, and the chain rule gives

$$(g \circ f)'(x_0) = 2\left(\frac{x_0 - 2}{x_0 - 1}\right)\frac{1}{(x_0 - 1)^2}.$$

Remark 10.1.17. If one writes y for $f(x)$, and z for $g(y)$, then the chain rule can be written in the more visually appealing manner $\frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx}$. However, this notation can be misleading (for instance it blurs the distinction between dependent variable and independent variable, especially for y), and leads one to believe that the quantities dz, dy, dx can be manipulated like real numbers. However, these quantities are not real numbers (in fact, we have not assigned any meaning to them at all), and treating them as such can lead to problems in the future. For instance, if f depends on x_1 and x_2, which depend on t, then chain rule for several variables asserts that $\frac{df}{dt} = \frac{\partial f}{\partial x_1}\frac{dx_1}{dt} + \frac{\partial f}{\partial x_2}\frac{dx_2}{dt}$, but this rule might seem suspect if one treated df, dt, etc. as real numbers. It is possible to think of dy, dx, etc. as "infinitesimal real numbers" if one knows what one is doing, but for those just starting out in analysis, I would not recommend this approach, especially if one wishes to work rigorously. (There is a way to make all of this rigorous, even for the calculus of several variables, but it requires the notion of a tangent

vector, and the derivative map, both of which are beyond the scope of this text.)

— Exercises —

Exercise 10.1.1. Suppose that X is a subset of $\mathbf{R}$, x_0 is a limit point of X, and $f : X \to \mathbf{R}$ is a function which is differentiable at x_0. Let $Y \subset X$ be such that $x_0 \in Y$, and x_0 is also a limit point of Y. Prove that the restricted function $f|_Y : Y \to \mathbf{R}$ is also differentiable at x_0, and has the same derivative as f at x_0. Explain why this does not contradict the discussion in Remark 10.1.2.

Exercise 10.1.2. Prove Proposition 10.1.7. (Hint: the cases $x = x_0$ and $x \neq x_0$ have to be treated separately.)

Exercise 10.1.3. Prove Proposition 10.1.10. (Hint: either use the limit laws (Proposition 9.3.14), or use Proposition 10.1.7.)

Exercise 10.1.4. Prove Theorem 10.1.13. (Hint: use the limit laws in Proposition 9.3.14. Use earlier parts of this theorem to prove the latter. For the product rule, use the identity

$$\begin{aligned} f(x)g(x) - f(x_0)g(x_0) \\ &= f(x)g(x) - f(x)g(x_0) + f(x)g(x_0) - f(x_0)g(x_0) \\ &= f(x)(g(x) - g(x_0)) + (f(x) - f(x_0))g(x_0); \end{aligned}$$

this trick of adding and subtracting an intermediate term is sometimes known as the "middle-man trick" and is very useful in analysis.)

Exercise 10.1.5. Let n be a natural number, and let $f : \mathbf{R} \to \mathbf{R}$ be the function $f(x) := x^n$. Show that f is differentiable on $\mathbf{R}$ and $f'(x) = nx^{n-1}$ for all $x \in \mathbf{R}$. (Hint: use Theorem 10.1.13 and induction.)

Exercise 10.1.6. Let n be a *negative* integer, and let $f : \mathbf{R} - \{0\} \to \mathbf{R}$ be the function $f(x) := x^n$. Show that f is differentiable on $\mathbf{R}$ and $f'(x) = nx^{n-1}$ for all $x \in \mathbf{R} - \{0\}$. (Hint: use Theorem 10.1.13 and Exercise 10.1.5.)

Exercise 10.1.7. Prove Theorem 10.1.15. (Hint: one way to do this is via Newton's approximation, Proposition 10.1.7. Another way is to use Proposition 9.3.9 and Proposition 10.1.10 to convert this problem into one involving limits of sequences, however with the latter strategy one has to treat the case $f'(x_0) = 0$ separately, as some division-by-zero subtleties can occur in that case.)

10.2 Local maxima, local minima, and derivatives

As you learnt in your basic calculus courses, one very common application of using derivatives is to locate maxima and minima. We now present this material again, but this time in a rigorous manner.

The notion of a function $f : X \to \mathbf{R}$ attaining a maximum or minimum at a point $x_0 \in X$ was defined in Definition 9.6.5. We now localize this definition:

Definition 10.2.1 (Local maxima and minima)**.** Let $f : X \to \mathbf{R}$ be a function, and let $x_0 \in X$. We say that f attains a *local maximum* at x_0 iff there exists a $\delta > 0$ such that the restriction $f|_{X\cap(x_0-\delta,x_0+\delta)}$ of f to $X \cap (x_0 - \delta, x_0 + \delta)$ attains a maximum at x_0. We say that f attains a *local minimum* at x_0 iff there exists a $\delta > 0$ such that the restriction $f|_{X\cap(x_0-\delta,x_0+\delta)}$ of f to $X \cap (x_0 - \delta, x_0 + \delta)$ attains a minimum at x_0.

Remark 10.2.2. If f attains a maximum at x_0, we sometimes say that f attains a *global* maximum at x_0, in order to distinguish it from the local maxima defined here. Note that if f attains a global maximum at x_0, then it certainly also attains a local maximum at this x_0, and similarly for minima.

Example 10.2.3. Let $f : \mathbf{R} \to \mathbf{R}$ denote the function $f(x) := x^2 - x^4$. This function does not attain a global minimum at 0, since for example $f(2) = -12 < 0 = f(0)$, however it does attain a local minimum, for if we choose $\delta := 1$ and restrict f to the interval $(-1, 1)$, then for all $x \in (-1, 1)$ we have $x^4 \leq x^2$ and thus $f(x) = x^2 - x^4 \geq 0 = f(0)$, and so $f|_{(-1,1)}$ has a local minimum at 0.

Example 10.2.4. Let $f : \mathbf{Z} \to \mathbf{R}$ be the function $f(x) = x$, defined on the integers only. Then f has no global maximum or global minimum (why?), but attains both a local maximum and local minimum at every integer n (why?).

Remark 10.2.5. If $f : X \to \mathbf{R}$ attains a local maximum at a point x_0 in X, and $Y \subset X$ is a subset of X which contains x_0, then the restriction $f|_Y : Y \to \mathbf{R}$ also attains a local maximum at x_0 (why?). Similarly for minima.

The connection between local maxima, minima and derivatives is the following.

Proposition 10.2.6 (Local extrema are stationary)**.** *Let $a < b$ be real numbers, and let $f : (a, b) \to \mathbf{R}$ be a function. If $x_0 \in (a, b)$, f is differ-*

entiable at x_0, and f attains either a local maximum or local minimum at x_0, then $f'(x_0) = 0$.

Proof. See Exercise 10.2.1. □

Note that f must be differentiable for this proposition to work; see Exercise 10.2.2. Also, this proposition does not work if the open interval (a, b) is replaced by a closed interval $[a, b]$. For instance, the function $f : [1, 2] \to \mathbf{R}$ defined by $f(x) := x$ has a local maximum at $x_0 = 2$ and a local minimum $x_0 = 1$ (in fact, these local extrema are global extrema), but at both points the derivative is $f'(x_0) = 1$, not $f'(x_0) = 0$. Thus the endpoints of an interval can be local maxima or minima even if the derivative is not zero there. Finally, the converse of this proposition is false (Exercise 10.2.3).

By combining Proposition 10.2.6 with the maximum principle, one can obtain

Theorem 10.2.7 (Rolle's theorem)**.** *Let $a < b$ be real numbers, and let $g : [a, b] \to \mathbf{R}$ be a continuous function which is differentiable on (a, b). Suppose also that $g(a) = g(b)$. Then there exists an $x \in (a, b)$ such that $g'(x) = 0$.*

Proof. See Exercise 10.2.4. □

Remark 10.2.8. Note that we only assume f is differentiable on the open interval (a, b), though of course the theorem also holds if we assume f is differentiable on the closed interval $[a, b]$, since this is larger than (a, b).

Rolle's theorem has an important corollary.

Corollary 10.2.9 (Mean value theorem)**.** *Let $a < b$ be real numbers, and let $f : [a, b] \to \mathbf{R}$ be a function which is continuous on $[a, b]$ and differentiable on (a, b). Then there exists an $x \in (a, b)$ such that $f'(x) = \frac{f(b)-f(a)}{b-a}$.*

Proof. See Exercise 10.2.5. □

— Exercises —

Exercise 10.2.1. Prove Proposition 10.2.6.

Exercise 10.2.2. Give an example of a function $f : (-1, 1) \to \mathbf{R}$ which is continuous and attains a global maximum at 0, but which is not differentiable at 0. Explain why this does not contradict Proposition 10.2.6.

Exercise 10.2.3. Give an example of a function $f : (-1, 1) \to \mathbf{R}$ which is differentiable, and whose derivative equals 0 at 0, but such that 0 is neither a local minimum nor a local maximum. Explain why this does not contradict Proposition 10.2.6.

Exercise 10.2.4. Prove Theorem 10.2.7. (Hint: use Corollary 10.1.12 and the maximum principle, Proposition 9.6.7, followed by Proposition 10.2.6. Note that the maximum principle does not tell you whether the maximum or minimum is in the open interval (a, b) or is one of the boundary points a, b, so you have to divide into cases and use the hypothesis $g(a) = g(b)$ somehow.)

Exercise 10.2.5. Use Theorem 10.2.7 to prove Corollary 10.2.9. (Hint: consider a function of the form $f(x) - cx$ for some carefully chosen real number c.)

Exercise 10.2.6. Let $M > 0$, and let $f : [a, b] \to \mathbf{R}$ be a function which is continuous on $[a, b]$ and differentiable on (a, b), and such that $|f'(x)| \leq M$ for all $x \in (a, b)$ (i.e., the derivative of f is bounded). Show that for any $x, y \in [a, b]$ we have the inequality $|f(x) - f(y)| \leq M|x - y|$. (Hint: apply the mean value theorem (Corollary 10.2.9) to a suitable restriction of f.) Functions which obey the bound $|f(x) - f(y)| \leq M|x - y|$ are known as *Lipschitz continuous functions* with *Lipschitz constant* M; thus this exercise shows that functions with bounded derivative are Lipschitz continuous.

Exercise 10.2.7. Let $f : \mathbf{R} \to \mathbf{R}$ be a differentiable function such that f' is bounded. Show that f is uniformly continuous. (Hint: use the preceding exercise.)

10.3 Monotone functions and derivatives

In your elementary calculus courses, you may have come across the assertion that a positive derivative meant an increasing function, and a negative derivative meant a decreasing function. This statement is not completely accurate, but it is pretty close; we now give the precise version of these statements below.

Proposition 10.3.1. *Let X be a subset of $\mathbf{R}$, let $x_0 \in X$ be a limit point of X, and let $f : X \to \mathbf{R}$ be a function. If f is monotone increasing* and *f is differentiable at x_0, then $f'(x_0) \geq 0$. If f is monotone decreasing* and *f is differentiable at x_0, then $f'(x_0) \leq 0$.*

Proof. See Exercise 10.3.1. □

Remark 10.3.2. We have to assume that f is differentiable at x_0; There exist monotone functions which are not always differentiable (see Exercise 10.3.2), and of course if f is not differentiable at x_0 we cannot possibly conclude that $f'(x_0) \geq 0$ or $f'(x_0) \leq 0$.

One might naively guess that if f were *strictly* monotone increasing, and f was differentiable at x_0, then the derivative $f'(x_0)$ would be strictly positive instead of merely non-negative. Unfortunately, this is not always the case (Exercise 10.3.3).

On the other hand, we do have a converse result: if function has strictly positive derivative, then it must be strictly monotone increasing:

Proposition 10.3.3. *Let $a < b$, and let $f : [a, b] \to \mathbf{R}$ be a differentiable function. If $f'(x) > 0$ for all $x \in [a, b]$, then f is strictly monotone increasing. If $f'(x) < 0$ for all $x \in [a, b]$, then f is strictly monotone decreasing. If $f'(x) = 0$ for all $x \in [a, b]$, then f is a constant function.*

Proof. See Exercise 10.3.4. □

— Exercises —

Exercise 10.3.1. Prove Proposition 10.3.1.

Exercise 10.3.2. Give an example of a function $f : (-1, 1) \to \mathbf{R}$ which is continuous and monotone increasing, but which is not differentiable at 0. Explain why this does not contradict Proposition 10.3.1.

Exercise 10.3.3. Give an example of a function $f : \mathbf{R} \to \mathbf{R}$ which is strictly monotone increasing and differentiable, but whose derivative at 0 is zero. Explain why this does not contradict Proposition 10.3.1 or Proposition 10.3.3. (Hint: look at Exercise 10.2.3.)

Exercise 10.3.4. Prove Proposition 10.3.3. (Hint: you do not have integrals or the fundamental theorem of calculus yet, so these tools cannot be used. However, one can proceed via the mean-value theorem, Corollary 10.2.9.)

Exercise 10.3.5. Give an example of a subset $X \subset \mathbf{R}$ and a function $f : X \to \mathbf{R}$ which is differentiable on X, is such that $f'(x) > 0$ for all $x \in X$, but f is not strictly monotone increasing. (Hint: the conditions here are subtly different from those in Proposition 10.3.3. What is the difference, and how can one exploit that difference to obtain the example?)

10.4 Inverse functions and derivatives

We now ask the following question: if we know that a function $f : X \to Y$ is differentiable, and it has an inverse $f^{-1} : Y \to X$, what can we say about the differentiability of f^{-1}? This will be useful for many applications, for instance if we want to differentiate the function $f(x) := x^{1/n}$.

We begin with a preliminary result.

Lemma 10.4.1. *Let $f : X \to Y$ be an invertible function, with inverse $f^{-1} : Y \to X$. Suppose that $x_0 \in X$ and $y_0 \in Y$ are such that $y_0 = f(x_0)$ (which also implies that $x_0 = f^{-1}(y_0)$). If f is differentiable at x_0,* and *f^{-1} is differentiable at y_0, then*

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Proof. From the chain rule (Theorem 10.1.15) we have

$$(f^{-1} \circ f)'(x_0) = (f^{-1})'(y_0) f'(x_0).$$

But $f^{-1} \circ f$ is the identity function on X, and hence by Theorem 10.1.13(b) $(f^{-1} \circ f)'(x_0) = 1$. The claim follows. □

As a particular corollary of Lemma 10.4.1, we see that if f is differentiable at x_0 with $f'(x_0) = 0$, then f^{-1} cannot be differentiable at $y_0 = f(x_0)$, since $1/f'(x_0)$ is undefined in that case. Thus for instance, the function $g : [0, \infty) \to [0, \infty)$ defined by $g(y) := y^{1/3}$ cannot be differentiable at 0, since this function is the inverse $g = f^{-1}$ of the function $f : [0, \infty) \to [0, \infty)$ defined by $f(x) := x^3$, and this function has a derivative of 0 at $f^{-1}(0) = 0$.

If one writes $y = f(x)$, so that $x = f^{-1}(y)$, then one can write the conclusion of Lemma 10.4.1 in the more appealing form $dx/dy = 1/(dy/dx)$. However, as mentioned before, this way of writing things, while very convenient and easy to remember, can be misleading and cause errors if applied too carelessly (especially when one begins to work in the calculus of several variables).

Lemma 10.4.1 seems to answer the question of how to differentiate the inverse of a function, however it has one significant drawback: the lemma only works if one assumes *a priori* that f^{-1} is differentiable. Thus, if one does not already know that f^{-1} is differentiable, one cannot use Lemma 10.4.1 to compute the derivative of f^{-1}.

However, the following improved version of Lemma 10.4.1 will compensate for this fact, by relaxing the requirement on f^{-1} from differentiability to continuity.

Theorem 10.4.2 (Inverse function theorem)**.** *Let $f : X \to Y$ be an invertible function, with inverse $f^{-1} : Y \to X$. Suppose that $x_0 \in X$*

and $y_0 \in Y$ are such that $f(x_0) = y_0$. If f is differentiable at x_0, f^{-1} is continuous at y_0, and $f'(x_0) \neq 0$, then f^{-1} is differentiable at y_0 and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Proof. We have to show that

$$\lim_{y \to y_0; y \in Y - \{y_0\}} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{1}{f'(x_0)}.$$

By Proposition 9.3.9, it suffices to show that

$$\lim_{n \to \infty} \frac{f^{-1}(y_n) - f^{-1}(y_0)}{y_n - y_0} = \frac{1}{f'(x_0)}$$

for any sequence $(y_n)_{n=1}^{\infty}$ of elements in $Y - \{y_0\}$ which converge to y_0.

To prove this, we set $x_n := f^{-1}(y_n)$. Then $(x_n)_{n=1}^{\infty}$ is a sequence of elements in $X - \{x_0\}$. (Why? Note that f^{-1} is a bijection) Since f^{-1} is continuous by assumption, we know that $x_n = f^{-1}(y_n)$ converges to $f^{-1}(y_0) = x_0$ as $n \to \infty$. Thus, since f is differentiable at x_0, we have (by Proposition 9.3.9 again)

$$\lim_{n \to \infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} = f'(x_0).$$

But since $x_n \neq x_0$ and f is a bijection, the fraction $\frac{f(x_n) - f(x_0)}{x_n - x_0}$ is non-zero. Also, by hypothesis $f'(x_0)$ is non-zero. So by limit laws

$$\lim_{n \to \infty} \frac{x_n - x_0}{f(x_n) - f(x_0)} = \frac{1}{f'(x_0)}.$$

But since $x_n = f^{-1}(y_n)$ and $x_0 = f^{-1}(y_0)$, we thus have

$$\lim_{n \to \infty} \frac{f^{-1}(y_n) - f^{-1}(y_0)}{y_n - y_0} = \frac{1}{f'(x_0)}$$

as desired. □

We give some applications of the inverse function theorem in the exercises below.

— Exercises —

Exercise 10.4.1. Let $n \geq 1$ be a natural number, and let $g : (0, \infty) \to (0, \infty)$ be the function $g(x) := x^{1/n}$.

(a) Show that g is continuous on $(0, \infty)$. (Hint: use Proposition 9.8.3.)

(b) Show that g is differentiable on $(0, \infty)$, and that $g'(x) = \frac{1}{n} x^{\frac{1}{n}-1}$ for all $x \in (0, \infty)$. (Hint: use the inverse function theorem and (a).)

Exercise 10.4.2. Let q be a rational number, and let $f : (0, \infty) \to \mathbf{R}$ be the function $f(x) = x^q$.

(a) Show that f is differentiable on $(0, \infty)$ and that $f'(x) = qx^{q-1}$. (Hint: use Exercise 10.4.1 and the laws of differential calculus in Theorem 10.1.13 and Theorem 10.1.15.)

(b) Show that $\lim_{x \to 1; x \in (0,\infty)} \frac{x^q - 1}{x - 1} = q$ for every rational number q. (Hint: use part (a) and Definition 10.1.1. An alternate route is to apply L'Hôpital's rule from the next section.)

Exercise 10.4.3. Let α be a real number, and let $f : (0, \infty) \to \mathbf{R}$ be the function $f(x) = x^\alpha$.

(a) Show that $\lim_{x \to 1; x \in (0,\infty) \setminus \{1\}} \frac{f(x) - f(1)}{x - 1} = \alpha$. (Hint: use Exercise 10.4.2 and the comparison principle; you may need to consider right and left limits separately. Proposition 5.4.14 may also be helpful.)

(b) Show that f is differentiable on $(0, \infty)$ and that $f'(x) = \alpha x^{\alpha-1}$. (Hint: use (a), exponent laws (Proposition 6.7.3), and Definition 10.1.1.)

10.5 L'Hôpital's rule

Finally, we present a version of a rule you are all familiar with.

Proposition 10.5.1 (L'Hôpital's rule I)**.** *Let X be a subset of $\mathbf{R}$, let $f : X \to \mathbf{R}$ and $g : X \to \mathbf{R}$ be functions, and let $x_0 \in X$ be a limit point of X. Suppose that $f(x_0) = g(x_0) = 0$, that f and g are both differentiable at x_0, but $g'(x_0) \neq 0$. Then there exists a $\delta > 0$ such that $g(x) \neq 0$ for all $x \in (X \cap (x_0 - \delta, x_0 + \delta)) - \{x_0\}$, and*

$$\lim_{x \to x_0; x \in (X \cap (x_0 - \delta, x_0 + \delta)) - \{x_0\}} \frac{f(x)}{g(x)} = \frac{f'(x_0)}{g'(x_0)}.$$

Proof. See Exercise 10.5.1. □

The presence of the δ here may seem somewhat strange, but is needed because $g(x)$ might vanish at some points other than x_0, which would

imply that quotient $\frac{f(x)}{g(x)}$ is not necessarily defined at all points in $X - \{x_0\}$.

A more sophisticated version of L'Hôpital's rule is the following.

Proposition 10.5.2 (L'Hôpital's rule II)**.** *Let $a < b$ be real numbers, let $f : [a, b] \to \mathbf{R}$ and $g : [a, b] \to \mathbf{R}$ be functions which are differentiable on $[a, b]$. Suppose that $f(a) = g(a) = 0$, that g' is non-zero on $[a, b]$ (i.e., $g'(x) \neq 0$ for all $x \in [a, b]$), and $\lim_{x \to a; x \in (a,b]} \frac{f'(x)}{g'(x)}$ exists and equals L. Then $g(x) \neq 0$ for all $x \in (a, b]$, and $\lim_{x \to a; x \in (a,b]} \frac{f(x)}{g(x)}$ exists and equals L.*

Remark 10.5.3. This proposition only considers limits to the right of a, but one can easily state and prove a similar proposition for limits to the left of a, or around both sides of a. Speaking very informally, the proposition states that

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

though one has to ensure all of the conditions of the proposition hold (in particular, that $f(a) = g(a) = 0$, and that the right-hand limit exists), before one can apply L'Hôpital's rule.

Proof. (Optional) We first show that $g(x) \neq 0$ for all $x \in (a, b]$. Suppose for sake of contradiction that $g(x) = 0$ for some $x \in (a, b]$. But since $g(a)$ is also zero, we can apply Rolle's theorem to obtain $g'(y) = 0$ for some $a < y < x$, but this contradicts the hypothesis that g' is non-zero on $[a, b]$.

Now we show that $\lim_{x \to a; x \in (a,b]} \frac{f(x)}{g(x)} = L$. By Proposition 9.3.9, it will suffice to show that

$$\lim_{n \to \infty} \frac{f(x_n)}{g(x_n)} = L$$

for any sequence $(x_n)_{n=1}^{\infty}$ taking values in $(a, b]$ which converges to x.

Consider a single x_n, and consider the function $h_n : [a, x_n] \to \mathbf{R}$ defined by

$$h_n(x) := f(x)g(x_n) - g(x)f(x_n).$$

Observe that h_n is continuous on $[a, x_n]$ and equals 0 at both a and x_n, and is differentiable on (a, x_n) with derivative $h_n'(x) = f'(x)g(x_n) - g'(x)f(x_n)$. (Note that $f(x_n)$ and $g(x_n)$ are constants with respect to

x.) By Rolle's theorem (Theorem 10.2.7), we can thus find $y_n \in (a, x_n)$ such that $h_n'(y_n) = 0$, which implies that

$$\frac{f(x_n)}{g(x_n)} = \frac{f'(y_n)}{g'(y_n)}.$$

Since $y_n \in (a, x_n)$ for all n, and x_n converges to a as $n \to \infty$, we see from the squeeze test (Corollary 6.4.14) that y_n also converges to a as $n \to \infty$. Thus $\frac{f'(y_n)}{g'(y_n)}$ converges to L, and thus $\frac{f(x_n)}{g(x_n)}$ also converges to L, as desired. $\square$

— Exercises —

Exercise 10.5.1. Prove Proposition 10.5.1. (Hint: to show that $g(x) \neq 0$ near x_0, you may wish to use Newton's approximation (Proposition 10.1.7). For the rest of the proposition, use limit laws, Proposition 9.3.14.)

Exercise 10.5.2. Explain why Exercise 1.2.12 does not contradict either of the propositions in this section.